

JULES LITMAN-CLEPER

Evolutionary Neuroscience | Biological Computational Cognition | Critical Simulation Studies

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RESEARCH PROFILE

Interdisciplinary researcher studying how sensory representations, memory systems, and biological information processing change across evolution, from cellular signaling to complex cognition. Combines computational modeling, comparative evolutionary neuroscience, and Science and Technology Studies to examine when similar biological capacities share information-processing principles, when they arise through distinct mechanisms, and what scientific models reveal or omit about those mechanisms.

Current work: Computational simulation of sensory-memory dynamics and developmental organization in *Physarum polycephalum*; Critical Simulation Studies of biological models.

Long-term goal: Develop comparative methods for characterizing computation across cellular, neural, collective, and artificial systems.

EDUCATION

University of California, Davis | Davis, CA 2025-present
PhD Candidate in Performance Studies; Designated Emphasis in Science and Technology Studies | Philosophy and Biological Computation

Focus: Critical Simulation Studies; Designated Emphasis in Science and Technology Studies; Advisor: Joe Dumit (STS)

Affiliation: Center for Artificial Intelligence and Experimental Futures (CAIEF)

University of California, Davis | Davis, CA 2022-2024

Master of Science in Psychology | Center for Neuroscience; Krubitzer Lab for Evolutionary Neurobiology

Thesis: *Cortical Field Complexity: A Comparative Analysis of Cortical Field Sizes in Relation to Ethology and Sensory Specialization Across Six Clades of Mammals*

Advisors: Leah Krubitzer (CNS), Stuart P. Wilson, and Randy O'Reilly (CNS)

Massachusetts Institute of Technology | Cambridge, MA 2014-2016

Master of Science in Architecture Studies (SMArchS) | Design and Computation and Brain and Cognitive Science

Thesis: *Situated Mappings: Augmented-Reality Clay and Adaptive Interfacing*

Advisors: Terry Knight (SMArchS); Ruth Rosenholz (CSAIL)

University of California, Davis | Davis, CA 2008-2011

Bachelor of Arts in Studio Art | Minor in Creative Writing; Science and Technology Studies

RESEARCH EXPERIENCE

University of California, Davis | Davis, CA 2025-present

Doctoral Researcher | *Biological Computation and Philosophy of Modeling*

- Models feedback between sensory information and memory, asking how these dynamics could contribute to the evolution of different forms of memory.
- Uses *Physarum polycephalum* to study changes in information processing as uninucleate amoebae develop into a multinucleate, networked plasmodium and then differentiate into multicellularity.
- Develops Critical Simulation Studies research on how model failures under perturbation can reveal mismatches between predictive outputs and underlying biological dynamics.

Krubitzer Lab for Evolutionary Neurobiology | UC Davis 2022-2024

Graduate Researcher | *Comparative Evolutionary Neuroscience*

- Conducted a cross-species analysis of cortical field size and complexity across six mammalian clades in relation to ethology and sensory specialization.
- Integrated comparative neurobiology, cortical organization, and evolutionary interpretation in the master's thesis project.
- Visited the Self-Organization Lab at the University of Sheffield and developed a multilayer perceptron (MLP) neural network to simulate cortical field organization. Studied the Baldwin effect with Stuart P. Wilson by reproducing Hinton and Nowlan's model from "How Learning Can Guide Evolution."

METHODS AND TECHNICAL SKILLS

Computational and quantitative: Blender; Blender Python; R and RStudio; Python; information theory; dynamical systems; generative algorithms

Interactive: Unity; C#; openFrameworks; machine vision; augmented and virtual reality; physical computing; embedded sensors; data sonification

OTHER RESEARCH AND TECHNICAL EXPERIENCE

- OpenAI - Projects Elephant and Lion | Remote** 2024
Biology Expert Trainer
- Developed expert questions, reference answers, and evaluation criteria grounded in evolutionary neuroscience and psychology.
 - Recognized as a top performer in the biology expert evaluation for Project Lion.
- MIT Design and Computation / BCS / CSAIL | Cambridge, MA** 2014-2016
Graduate Researcher
- Designed an augmented-reality clay interface to investigate adaptive mappings between physical manipulation and digital representation.
- Nokia Bell Labs | Sunnyvale, CA** 2020
Artist in Residence, Research and Development
- Investigated ecological communication systems through the project *Earth-Centered Communication Technology*.
- Universe Sandbox | San Francisco, CA** 2018
Sound Designer
- Created data-driven sonification of space-physics simulations using Unity and C#.
- Windchimes Electronics | Richmond, CA** 2017
Machine Vision Specialist
- Implemented Eulerian video magnification using openFrameworks.
- Virtual Reality Installation, Historical Museum of New Orleans | New Orleans, LA** 2016
Motion-Capture Workflow Consultant
- Supported a Unity- and Kinect-based virtual-reality installation workflow.
- Exploratorium | San Francisco, CA** 2011-2014
Tactile Dome Specialist and Exhibit Technician
- Prototyped touch-based interactive installations, conducted visitor research and survey analysis, and supported exhibit fabrication and maintenance.

SELECTED TEACHING

- University of San Francisco | San Francisco, CA** 2023
Lecturer in Art and Architecture | *Interaction Design*
- San Francisco State University | San Francisco, CA** 2021
Lecturer of New Media and Interactive Arts | *Physical Computing with Embedded Sensors*
- California State University, East Bay | Hayward, CA** 2019-2021
Adjunct Professor of Multimedia Art
- Taught interaction design, 3D foundations and digital fabrication, time-based media, and graduate experiments in art and technology.
- Massachusetts Institute of Technology | Cambridge, MA** 2016
Teaching Assistant and Lead Teacher
- Supported rapid prototyping, urban energy-flow modeling, CNC fabrication, architecture studio, and an augmented-reality and dance course.

SELECTED HONORS AND AWARDS

Massachusetts Institute of Technology: Merit Scholarship; Louis C. Rosenberg Travel Fellowship
University of California, Davis: Dean's Honors List (three times); Wayne Thiebaud Family Scholarship (twice); New York Studio School Scholarship; Yale/Norfolk nomination; Nettie Weber Scholarship
Augmented Reality World Expo: Award for EmForm, a browser-connected, augmented-reality collective sculpting project

CURRICULUM VITAE

PUBLICATIONS

- 2026 Litman-Cleper, Jules. "Sensing Is Becoming: How a Rootless Parasitic Plant Blossoms in Spring." *Bay Nature*, Spring 2026, March 21, 2026.
- 2024 Litman-Cleper, Jules. *Cortical Field Complexity: Comparative Analysis of Cortical Field Sizes in Relation to Ethology and Sensory Specialization across 6 Clades of Mammals*. M.S. thesis, University of California, Davis, 2024.
- 2024 Bresee, Chris, et al. "Translating the Timing of Developmental Benchmarks in Short-Tailed Opossums (*Monodelphis domestica*) to Facilitate Comparisons with Commonly Used Rodent Models." *Brain, Behavior and Evolution* 99, no. 2 (2024): 69-85. <https://doi.org/10.1159/000538524>.
- 2023 Dumit, Joseph, and Jules Litman-Cleper. "FAB! Fungal-Algae-Bacteria ~ Lichen: Fabulous, Fabulating, Falling into Place, Folding Ambient Biofilms." *Tumbling Ecologies*.
- 2020 Litman-Cleper, Julia. "Earth-Centered Communication Technologies." *Leonardo Journal of Art and Science*.
- 2016 Litman-Cleper, Julia. *Situated Mappings: Augmented Reality Clay and Adaptive Interfacing*. S.M. thesis, Massachusetts Institute of Technology, Department of Architecture, 2016.
- 2015 Zaman, Cagri Hakan, and Julia Litman-Cleper. "Space First: A Spatial Inference Model for Multisensory Perception." Conference paper in *Space and Situated Cognition: Proceedings of the Sixth International Conference on Spatial Cognition (ICSC 2015)*, Rome, September 7-11, 2015. ISSN 1612-4790.

TALKS

- 2025 "Simple Geometry Proximity Sensing Using Blender Geometry Nodes to Simulate Biological Interdependency." Simulation Sessions, Gray Area Center for the Arts, CA.
- 2025 "Chaotic (Messy), Embedded (Alive), Multimodal (Sensory) Mappings (Training)." Lecture on live audiovisual performance, Center for Computer Research in Music and Acoustics (CCRMA), Stanford, CA.
- 2023 "The Organization of Cortical Patterns in the Context of Evolution and Experience." Biological Psychology Research Seminar brown bag talk, UC Davis, CA.
- 2020 "Unsimulatableness: Natural Computation and Ecological Encountering." Leonardo Art Science Rendezvous, University of San Francisco, San Francisco, CA.
- 2019 "Earth-Centered Communication Technology." Lecture at Nokia Bell Labs, Sunnyvale, CA.
- 2018 "Queering Generativity, Querying Generativity: On Generative Algorithms." Algorithmic Art Assembly, Gray Area Center for the Arts, San Francisco, CA.
- 2015 "Statistics and Space: How the Mind Encodes Spatial Dimensions." Cultural Computation, MIT, Cambridge, MA.
- 2015 "Sound Mappings: Inhabiting Virtual Geometry Through Sound." Center for Computer Research in Music and Acoustics (CCRMA), Palo Alto, CA.
- 2015 "Space First: A Spatial Inference Model for Multisensory Perception." Space and Situated Cognition: Proceedings of the Sixth International Conference on Spatial Cognition (ICSC 2015), Rome, Italy.
- 2013 "EmForm: Sculpting in Augmented Reality." Augmented Reality World Faire Hackathon, Mountain View, CA.

SELECTED EXHIBITIONS

Video, multimedia installation, and sculpture

- 2021 *TerraCycle, In Gratitude for Earth*. Co-curator and contributor of "A Meditation on Earthworms." Digital broadcast.
- 2019 Exhibition with CODAME through the Bell Labs Residency Program and *Leonardo* magazine.
- 2017 Gray Area Foundation for the Arts and Technology, *Vital Hybrids: HowToBuildABody*, San Francisco, CA.
- 2017 B4B3L4B Gallery, *Vital Hybrids: 3 Cross Sections*, Oakland, CA.
- 2017 Maker Faire, exhibition of augmented-reality clay, San Mateo, CA.
- 2015 ISEA, "Imagining Augmented Biological Architectures," interface installation, Vancouver, Canada.

SELECTED PERFORMANCES

Electronic music and interactive multimedia installation

- 2025 Shatter Pattern at the Center for Computer Research in Music and Acoustics (CCRMA), Stanford, CA.
- 2025 Shatter Pattern at Indexical, Santa Cruz, CA.
- 2025 Shatter Pattern at The Lab, San Francisco, CA.
- 2019 Algorithmic Art Assembly, Gray Area Center for the Arts and Technology, San Francisco, CA.
- 2017 San Francisco Electronic Music Festival (SFEMF), Brava Theatre, CA.

MUSIC RELEASES

- 2020 *Shatter Pattern Tape*, Embalming Lately, New York City.
- 2019 *Waxy Tomb* EP, Gilgongo Records, Arizona.
- 2013 "Infra-shape," Weird Ear Records, Oakland, CA.